

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250274145

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Takano; Atsushi et al.

PILLARLESS ANTENNA STRUCTURE FOR BULK ACOUSTIC WAVE FILTER

Abstract

A duplexer including a support substrate, a reception signal terminal, a transmit signal terminal, an antenna terminal, a receive side filter formed on the support substrate and electrically coupling the reception signal terminal to the antenna terminal, a transmit side filter formed on the support substrate and electrically coupling the transmit signal terminal to the antenna terminal, a side wall surrounding the receive side filter and the transmit side filter, and a cap wafer disposed on the side wall and enclosing the receive side filter and the transmit side filter in a cavity defined by the cap wafer, the side wall, and the support substrate, there being no post disposed on the antenna terminal and in contact with the cap wafer.

Inventors: Takano; Atsushi (Kadoma-Shi, JP), Komatsu; Tomoya (Irvine, CA), Gheethan; Ahmad Adel Ahmad (Newbury Park, CA), Sadasivan; Kshitij Sandeep (Sunnyvale, CA)

Applicant: SKYWORKS SOLUTIONS, INC. (Irvine, CA)

Family ID: 1000008463093

Appl. No.: 19/050355

Filed: February 11, 2025

Related U.S. Application Data

us-provisional-application US 63558432 20240227

Publication Classification

Int. Cl.: H04B1/00 (20060101); H03H9/70 (20060101); H04B1/40 (20150101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 63/558,432, titled “PILLARLESS ANTENNA STRUCTURE FOR BULK ACOUSTIC WAVE FILTER,” filed Feb. 27, 2024, the entire content of which is incorporated herein by reference for all purposes.

BACKGROUND

[0002] Conventionally, in communication devices such as mobile phones, filtering devices are used to separate signals having different bands, such as a transmission signal and a reception signal, or transmission or reception signals, in different bands, etc. Bulk acoustic wave (BAW) resonators have been used for such filtering devices. The BAW resonators may include film bulk acoustic resonators and/or solidly mounted resonators (SMRs).

[0003] A plurality of BAW resonators may be electrically connected to form a duplexer **10** in which both transmit circuitry **20** including a transmit signal terminal **30** and receive circuitry **40** including a receive signal terminal **50** are electrically coupled to an antenna **60** as schematically illustrated in FIG. **1**.

[0004] In some instances both the receive circuitry and the transmit circuitry of a duplexer are provided on a single die and formed on the same substrate. FIG. **2** is a simplified illustration of the arrangement of BAW resonators in one example of a duplexer package **100** in which the transmit and receive circuitry are in the form of ladder filters.

[0005] In the duplexer package **100**, the receive circuitry includes series-connected BAW resonators **111**, **113**, **115**, **117** disposed in series along a conductive signal path **131** extending from an antenna terminal **310** to a reception signal terminal **143** on the top surface region **420'** of a support substrate **420**. Parallel-connected BAW resonators **112**, **114**, **116** are connected between a ground plane **132** and respective nodes interconnecting the series-connected BAW resonators **111**, **113**, **115**, **117** along the conductive signal path **131**. The ground plane **132** connects the parallel-connected BAW resonators **112**, **114**, **116** to ground contacts/terminals **142**, **144**, **146**.

[0006] In the duplexer package **100**, the transmit circuitry includes series-connected BAW resonators **211**, **213**, **215**, **217** disposed in series along a conductive signal path **231** extending from the antenna terminal **310** to a transmit signal terminal **243** on the top surface region **420''** of the support substrate **420**. Parallel-connected BAW resonators **212**, **214**, **216** are connected between a ground plane **232** and respective nodes interconnecting the series-connected BAW resonators **211**, **213**, **215**, **217** along the conductive signal path **231**. The ground plane **232** connects the parallel-connected BAW resonators **212**, **214**, **216** to ground contacts/terminals **242**, **244**, **246**.

[0007] The duplexer package **100** includes a cap wafer **320** (shown in FIGS. **3-7**), which may be formed from Si and is disposed to oppose the top surface regions **420'**, **420''** of the support substrate **420** via a certain gap. A metal side wall **154** formed along the outer periphery on the top surface regions **420'**, **420''** of the support substrate **420** spaces the cap wafer **320** apart from the support substrate **420** and encloses the gap between the support substrate **420** and the cap wafer **320**. In some embodiments the side wall **154** may be grounded.

SUMMARY

[0008] In accordance with one aspect, there is provided a duplexer. The duplexer comprises a support substrate, a reception signal terminal, a transmit signal terminal, an antenna terminal, a receive side filter formed on the support substrate and electrically coupling the reception signal terminal to the antenna terminal, a transmit side filter formed on the support substrate and electrically coupling the transmit signal terminal to the antenna terminal, a side wall surrounding

the receive side filter and the transmit side filter, and a cap wafer disposed on the side wall and enclosing the receive side filter and the transmit side filter in a cavity defined by the cap wafer, the side wall, and the support substrate, a space being defined between the cap wafer and upper surface of the antenna terminal, the space being formed by an absence of a post disposed on the antenna terminal and in contact with the cap wafer.

[0009] In some embodiments, the duplexer further comprises an external antenna contact disposed on a side of the support substrate opposite the antenna terminal.

[0010] In some embodiments, the duplexer further comprises a through substrate via passing through the support substrate and electrically connecting the antenna terminal to the external antenna contact.

[0011] In some embodiments, the cap wafer is formed of silicon.

[0012] In some embodiments, a side of the cap wafer facing the support substrate include a trap-rich layer.

[0013] In some embodiments, the antenna terminal includes an adhesion layer, a first metal layer disposed on the adhesion layer, and a second metal layer disposed on the first metal layer.

[0014] In some embodiments, the second metal layer is formed of a different metal than the first metal layer.

[0015] In some embodiments, the second metal layer is at least as thick as the first metal layer.

[0016] In some embodiments, the receive side filter and the transmit side filter both include bulk acoustic wave resonators.

[0017] In some embodiments, the duplexer further comprises a post disposed on the reception signal terminal extending from the reception signal terminal to the cap wafer.

[0018] In some embodiments, the duplexer further comprises a post disposed on the transmit signal terminal and extending from the transmit signal terminal to the cap wafer.

[0019] In some embodiments, the duplexer further comprises one or more ground terminals disposed on the support substrate.

[0020] In some embodiments, the duplexer further comprises posts disposed on the one or more ground terminals and extending from the one or more ground terminals to the cap wafer.

[0021] In some embodiments, the duplexer is included in a front end module.

[0022] In some embodiments, the duplexer is included in a front end module that is included in an electronic device.

[0023] In accordance with another aspect, there is provided a method of forming a duplexer. The method comprises depositing a layer of metal including portions defining a reception signal terminal, a transmit signal terminal, and an antenna terminal on a support substrate, forming a receive side filter on the support substrate electrically coupling the reception signal terminal to the antenna terminal, forming a transmit side filter on the support substrate electrically coupling the transmit signal terminal to the antenna terminal, forming a metal side wall on the support substrate surrounding the receive side filter and the transmit side filter, and attaching a cap wafer to a top of the side wall, there being no post disposed on the antenna terminal and in contact with the cap wafer.

[0024] In some embodiments, the method further comprises forming the cap wafer from Si with a trap-rich layer on a lower side of the cap wafer.

[0025] In some embodiments, the method further comprises forming the antenna terminal from an adhesion layer, a first metal layer disposed on the adhesion layer, and a second metal layer disposed on the first metal layer.

[0026] In some embodiments, the method further comprises forming a post on the reception signal terminal extending from the reception signal terminal to the cap wafer.

[0027] In some embodiments, the method further comprises forming a post on the transmit signal terminal and extending from the transmit signal terminal to the cap wafer.

[0028] In some embodiments, the method further comprises forming one or more ground terminals

on the support substrate with posts disposed on the one or more ground terminals and extending from the one or more ground terminals to the cap wafer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of the invention. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

[0030] FIG. 1 is a circuit diagram showing a configuration of a duplexer;

[0031] FIG. 2 is a plan view of a duplexer package including a ladder-type transmit and receive filters formed using BAW resonators;

[0032] FIG. 3 is a cross-sectional diagram of a portion of an example of a duplexer including an antenna terminal and a sidewall;

[0033] FIG. 4 illustrates a modification to the duplexer of FIG. 3 in which a portion of an adhesion layer between the antenna terminal and sidewall has been omitted;

[0034] FIG. 5 illustrates a modification to the duplexer of FIG. 3 in which a trap-rich region has been added to the cap wafer;

[0035] FIG. 6 illustrates a modification to the duplexer of FIG. 3 in which a post disposed on the antenna terminal and extending between the antenna terminal and the cap wafer has been removed;

[0036] FIG. 7 illustrates a modification to the duplexer of FIG. 3 in which both a portion of an adhesion layer between the antenna terminal and sidewall has been omitted and in which a post disposed on the antenna terminal and extending between the antenna terminal and the cap wafer has been removed;

[0037] FIG. 8 illustrates results of simulations of parasitic signal coupling between the antenna terminal and receive terminal in examples of duplexers having the structures illustrated in FIGS. 3-7.

[0038] FIG. 9 is block diagram of one example of a front end module in which a duplexer according to certain embodiments can be used; and

[0039] FIG. 10 is a block diagram of one example of a wireless device in which a duplexer according to certain embodiments can be used.

DETAILED DESCRIPTION

[0040] The following description of certain embodiments presents various descriptions of specific embodiments. However, the innovations described herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.

[0041] In a duplexer, signals within the transmit and receive circuitry are ideally fully isolated from one another so that, for example, signals input to the transmit terminal do not travel to the receive terminal. If too much transmit signal power reaches the receive terminal it could saturate a low noise amplifier coupled to the receive terminal or otherwise interfere with the receive side circuitry.

[0042] Transmit signals may travel from an antenna terminal 310 or a transmit terminal 243 of a

duplexer to a receive terminal **143** of the duplexer by coupling through, for example, the side wall **154** and/or through a cap wafer **320**. A post **410** that is present in many duplexer designs that extends from the antenna terminal **310** to an inner lower surface of the cap wafer **320** may provide a path for signal coupling from the antenna terminal **310** to the cap wafer **320**. Signals coupled from the antenna terminal **310** to the cap wafer **320** may travel through the cap wafer **320** to other portions of the duplexer, for example, the receive terminal **143**.

[0043] In the example of FIGS. 3-7, an external antenna contact **360** is on the outside of the device or support substrate **420** and coupled to the antenna terminal **310** by a through substrate via (TSV) **430** that passes through the device or support substrate **420**. The device or support substrate **420** may be formed of a lower layer **420a** of Si, a layer **420b** of trap-rich Si, and an upper layer **420c** of SiO.sub.2. An upper end of the TSV **430** terminates on the antenna terminal **310** which is formed from a first metal layer **440** (e.g., Mo, Ru, or W) and a metal 1 layer **450** (e.g., Cu, Au, or Ti/Au alloy). Other portions of these metal layers **440**, **450** are used to form portions of resonators in the duplexer (not shown in FIGS. 3-7.) The side wall **154** also extends from portions of the metal layers **440**, **450** atop the device or support substrate **420** to the lower internal surface of the cap wafer **320**. In some embodiments, portions of an adhesion layer **460** formed of, for example, Sc-doped AlN and a silicon dioxide layer **470**, other portions of which are used to form portions of resonators in the duplexer, may be disposed on the device or support substrate **420** between the antenna terminal **310** and the side wall **154**. Without wishing to be bound to a particular theory, it is believed that in the process of depositing the adhesion layer **460** and SiO.sub.2 layer **470** between the antenna terminal **310** and the side wall **154**, some form of conductive residue or fixed charges may be introduced onto the device or support substrate **420** between the antenna terminal **310** and the side wall **154**. This residue or fixed charges may facilitate coupling of signals from the antenna terminal to the side wall **154** which may then travel through the side wall **154** and cap wafer **320** and couple to the reception signal terminal **143** of the duplexer.

[0044] Accordingly, one way of reducing transmit signal coupling between the antenna terminal **310** or a transmit terminal **243** of a duplexer to a receive terminal **143** of the duplexer through the side wall **154** and/or cap wafer **320** may be to remove the portions of the adhesion layer **460** and oxide layer **470** between the antenna terminal **310** and sidewall **154**, for example, by etching, as illustrated in FIG. 4. Alternatively, one may utilize a fabrication process that does not include forming the portions of the adhesion layer **460** and oxide layer **470** on the device or support substrate **420** between the antenna terminal **310** and the side wall **154**. In some embodiments, unnecessary portions of the adhesion layer **460** and oxide layer **470** remaining between any conductive signal-carrying structures of the duplexer **100** and the sidewall **154**, for example, resonators or signal lines, could be removed (or not formed in the first place) to reduce the potential for coupling between the conductive signal-carrying structures and the sidewall **154**.

[0045] A method of further reducing transmit signal coupling between antenna terminal **310** or a transmit terminal **243** of a duplexer to a receive terminal **143** of the duplexer through the cap wafer **320** may be to increase the electrical resistance of the cap wafer **320**, or at least portions contacted by the post **410** or other posts within the duplexer. This may be accomplished by forming a trap-rich layer **325** in the lower portion of the cap wafer **320** that is contacted by the post **410**. An example of this is shown in FIG. 5.

[0046] In another embodiment, reducing transmit signal coupling between antenna terminal **310** of a duplexer to a receive terminal **143** of the duplexer through the post **410** and the cap wafer **320** may be to remove the post **410**, or not form the post **410** at all. The post **410** may provide mechanical support to the device or support substrate **420** and/or the cap wafer **320** so omission or removal of the post **410** may weaken the device or support substrate **420** and/or the cap wafer **320**. Removal or omission of the post **410** may thus increase risk of cracking of the cap wafer **320** or device or support substrate **420** due to mechanical stress generated during manufacturing at the pick-up and transfer process from dicing tape to emboss tape or from the emboss tape to mount on

a PCB. Removal or omission of the post **410** may additionally or alternatively allow for the device or support substrate **420** to undesirably flex upward toward the cap wafer **320**, potentially causing a short circuit or may allow for a mismatch in thermal expansion coefficient between the TSV **430** and materials of the device or support substrate **420** to cause the device or support substrate **420** to crack. Accordingly, in some embodiments in which the post **410** is removed or is not present, an additional metal layer **460** may be provided on top of or as a portion of the antenna terminal **310** to add additional strength to the device or support substrate **420** in the region where the post **410** would otherwise have been disposed. An example of this is shown in FIG. **6**. The additional metal layer **460** may be formed from the metal 2 layer of the duplexer **100** and may be as thick as or thicker than the metal 1 layer **450**. The metal 1 layer **450** may be, for example, a 0.6-0.7 μm thick layer of a Ti/Au alloy and the additional metal layer **460** may be a 2 μm thick layer of copper, although aspects and embodiments disclosed herein are not limited to metal layers formed of these materials or with these thicknesses.

[0047] In some embodiments there is no post **410** disposed on the antenna terminal **310**, although posts similar to post **410** may be disposed on any or all of the transmit terminal **243**, receive terminal **143**, or ground contacts/terminals **142**, **144**, **146**, **242**, **244**, **246** to enhance the mechanical strength of the duplexer module **100**.

[0048] Any of the structural modifications to the duplexer illustrated in FIGS. **4-6** may be combined. For example, one may remove or omit both the portions of the adhesion layer **460**, layer, and the post **410** and add the additional metal layer **460**. An example of this is illustrated in FIG. **7**.

[0049] Simulations were performed to determine the effect of the structural modifications to a duplexer as illustrated in FIGS. **4-7** on parasitic coupling between the antenna terminal of the duplexer and the receive terminal, specifically parasitic coupling of the third harmonic (H3) of the transmit signal in a duplexer operating at the WiFi 2.4 band. FIG. **8** illustrates results of these simulations. As shown in FIG. **8** a duplexer with the structure shown in FIG. **3** including the portions of the adhesion layer **460** and SiO.sub.2 layer **470** with a Si cap wafer (no trap-rich layer) **320**, and including the post **410** on the antenna terminal (the “With ADL/Si Cap/Ant Post” test results) exhibited a parasitic coupling of the H3 signal between the antenna terminal **310** and receive terminal with a signal strength of about -90 db. In the embodiment illustrated in FIG. **4** in which the portions of the adhesion layer **460** and SiO.sub.2 layer between the antenna terminal **310** and sidewall **154** was removed (the “Without ADL/Si Cap/Ant Post” test results) the parasitic coupling of the H3 signal between the antenna terminal **310** and receive terminal had a signal strength of about -100 db. In both the embodiment of FIG. **5** in which the cap wafer **320** included a trap-rich layer **325** (the “With ADL/Trap-Rich Cap/Ant Post” test results) and the embodiment of FIG. **6** in which the antenna terminal post **410** was omitted (the “With ADL/Si Cap/No Ant Post” test results) the parasitic coupling of the H3 signal between the antenna terminal **310** and receive terminal had a signal strength of about -110 db. In the embodiment of FIG. **7** in which the portions of the adhesion layer **460** and SiO.sub.2 layer **470** and the antenna terminal post **410** were omitted (the “Without ADL/Si Cap/No Ant Post” test results) the parasitic coupling of the H3 signal between the antenna terminal **310** and receive terminal had a signal strength of about -115 db. These results show that the most effective manner to reduce parasitic coupling of transmit signals from an antenna terminal to a receive terminal through the cap wafer in a duplexer as disclosed herein operating in the WiFi 2.4 band would be to omit each of the portions of the adhesion layer **460** and SiO.sub.2 layer identified above and the antenna terminal post **410**.

[0050] Embodiments of the duplexer **100** can be used in a wide variety of electronic devices and components. Referring to FIG. **9** there is illustrated a block diagram of one example of a front-end module **600**, which may be used in an electronic device such as a wireless communications device (e.g., a mobile phone) for example, and which includes an antenna duplexer **100** as disclosed herein. The antenna duplexer **100** has a common terminal **502**, an input terminal **504**, and an output terminal **506**. The antenna duplexer **100** may include one or more transmission filters **510**

connected between the input terminal **504** and the common terminal **502**, and one or more reception filters **520** connected between the common terminal **502** and the output terminal **506**. An antenna **710** is connected to the common terminal **502**. In certain examples, a phase matching component **610**, such as an inductor, can be connected to the common terminal **502**, as shown. The front-end module **600** further includes a transmitter circuit **622** connected to the input terminal **504** of the duplexer **100** and a receiver circuit **624** connected to the output terminal **506** of the duplexer **100**. The transmitter circuit **622** can generate signals for transmission via the antenna **710**, and the receiver circuit **624** can receive and process signals received via the antenna **710**. In some embodiments such reception and transmission functionalities can be implemented in separate components, as illustrated in FIG. **9**, or in a common transceiver circuit/module, as discussed further below. As will be appreciated by those skilled in the art, given the benefit of this disclosure, the front-end module **600** may include other components, not illustrated, such as, but not limited to, switches, electromagnetic couplers, amplifiers, processors, and the like.

[0051] Embodiments of the antenna duplexer **100** or front end module **600** may be advantageously used in a variety of electronic devices. Examples of the electronic devices can include, but are not limited to, consumer electronic products, parts of the consumer electronic products, electronic test equipment, cellular communications infrastructure such as a base station, etc. Examples of the electronic devices can include, but are not limited to, a mobile phone such as a smart phone, a telephone, a television, a computer monitor, a computer, a modem, a hand held computer, a laptop computer, a tablet computer, an electronic book reader, a wearable computer such as a smart watch, a personal digital assistant (PDA), an appliance, an automobile, a stereo system, a DVD player, a CD player, a digital music player such as an MP3 player, a radio, a camcorder, a camera, a digital camera, a portable memory chip, a health care monitoring device, a vehicular electronics system such as an automotive electronics system or an avionics electronic system, a peripheral device, a wrist watch, a clock, etc. Further, the electronic devices can include unfinished products.

[0052] FIG. **10** is a block diagram of one example of a wireless device **700** including an antenna duplexer **100** as disclosed herein. The wireless device **700** can be a cellular phone, smart phone, tablet, modem, communication network or any other portable or non-portable device configured for voice or data communication. The wireless device **700** can receive and transmit signals from the antenna **710**. The wireless device **700** includes the front-end module **600**, which includes the duplexer **100** as discussed above. The front-end module **600** further includes an antenna switch **630**, which can be configured to switch between different frequency bands or modes, such as transmit and receive modes, for example. In the example illustrated in FIG. **10**, the antenna switch **630** is positioned between the duplexer **100** and the antenna **710**; however, in other examples the duplexer **100** can be positioned between the antenna switch **630** and the antenna **710**, or the antenna switch **630** and the duplexer **100** can be integrated into a single module.

[0053] The front end module **600** includes a transceiver **620** that is configured to generate signals for transmission or to process received signals. The transceiver **620** can include the transmitter circuit **622** which can be connected to the input terminal **504** of the duplexer **100**, and the receiver circuit **624** which can be connected to the output terminal **506** of the duplexer **100**, as shown in FIG. **9**. Signals generated for transmission by the transmitter circuit **622** are received by a power amplifier (PA) module **640**, which amplifies the generated signals from the transceiver **620**. As will be appreciated by those skilled in the art, the power amplifier module **640** can include one or more power amplifiers. The power amplifier module **640** can be used to amplify a wide variety of RF or other frequency-band transmission signals. For example, the power amplifier module **640** can receive an enable signal that can be used to pulse the output of the power amplifier to aid in transmitting a wireless local area network (WLAN) signal or any other suitable pulsed signal. The power amplifier module **640** can be configured to amplify any of a variety of types of signal, including, for example, a Global System for Mobile (GSM) signal, a code division multiple access (CDMA) signal, a W-CDMA signal, a Long Term Evolution (LTE) signal, or an EDGE signal. In

certain embodiments, the power amplifier module **640** and associated components including switches and the like can be fabricated on GaAs substrates using, for example, pHEMT or BiFET transistors, or on a Silicon substrate using CMOS transistors. The front-end module **600** further includes a low noise amplifier module **650**, which amplifies received signals from the antenna **710** and provides the amplified signals to the receiver circuit **624** of the transceiver **620**.

[0054] The wireless device **700** of FIG. **10** further includes a power management sub-system **720** that is connected to the transceiver **620** and that manages the power for the operation of the wireless device. The power management system **720** can also control the operation of a baseband sub-system **730** and other components of the wireless device **700**. The power management system **720** can include, or can be connected to, a battery (not shown) that supplies power for the various components of the wireless device **700**. The power management system **720** can further include one or more processors or controllers that can control the transmission of signals, for example.

[0055] In one embodiment, the baseband sub-system **730** is connected to a user interface **740** to facilitate various input and output of voice or data provided to and received from the user. The baseband sub-system **730** can also be connected to a memory **750** that is configured to store data or instructions to facilitate the operation of the wireless device **700**, or to provide storage of information for the user.

[0056] Having described above several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this disclosure and are intended to be within the scope of the invention. Accordingly, it is to be appreciated that embodiments discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the description or illustrated in the accompanying drawings. The devices disclosed herein are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use herein of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. Accordingly, the foregoing description and drawings are by way of example only, and the scope of the invention should be determined from proper construction of the appended claims, and their equivalents.

Claims

1. A duplexer comprising: a support substrate; a reception signal terminal; a transmit signal terminal; an antenna terminal; a receive side filter formed on the support substrate and electrically coupling the reception signal terminal to the antenna terminal; a transmit side filter formed on the support substrate and electrically coupling the transmit signal terminal to the antenna terminal; a side wall surrounding the receive side filter and the transmit side filter; and a cap wafer disposed on the side wall and enclosing the receive side filter and the transmit side filter in a cavity defined by the cap wafer, the side wall, and the support substrate, a space being defined between the cap wafer and upper surface of the antenna terminal, the space being formed by an absence of a post disposed on the antenna terminal and in contact with the cap wafer.
2. The duplexer of claim 1 further comprising an external antenna contact disposed on a side of the support substrate opposite the antenna terminal.
3. The duplexer of claim 2 further comprising a through substrate via passing through the support substrate and electrically connecting the antenna terminal to the external antenna contact.

4. The duplexer of claim 1 wherein the cap wafer is formed of silicon.
 5. The duplexer of claim 4 wherein a side of the cap wafer facing the support substrate include a trap-rich layer.
 6. The duplexer of claim 1 wherein the antenna terminal includes an adhesion layer, a first metal layer disposed on the adhesion layer, and a second metal layer disposed on the first metal layer.
 7. The duplexer of claim 6 wherein the second metal layer is formed of a different metal than the first metal layer.
 8. The duplexer of claim 6 wherein the second metal layer is at least as thick as the first metal layer.
 9. The duplexer of claim 1 wherein the receive side filter and the transmit side filter both include bulk acoustic wave resonators.
 10. The antenna duplexer of claim 1 further comprising a post disposed on the reception signal terminal extending from the reception signal terminal to the cap wafer.
 11. The antenna duplexer of claim 10 further comprising a post disposed on the transmit signal terminal and extending from the transmit signal terminal to the cap wafer.
 12. The antenna duplexer of claim 11 further comprising one or more ground terminals disposed on the support substrate.
 13. The antenna duplexer of claim 12 further comprising posts disposed on the one or more ground terminals and extending from the one or more ground terminals to the cap wafer.
 14. A front end module including the duplexer of claim 1.
 15. An electronic device including the front end module of claim 14.
 16. A method of forming a duplexer, the method comprising: depositing a layer of metal including portions defining a reception signal terminal, a transmit signal terminal, and an antenna terminal on a support substrate; forming a receive side filter on the support substrate electrically coupling the reception signal terminal to the antenna terminal; forming a transmit side filter on the support substrate electrically coupling the transmit signal terminal to the antenna terminal; forming a metal side wall on the support substrate surrounding the receive side filter and the transmit side filter; and attaching a cap wafer to a top of the side wall, there being no post disposed on the antenna terminal and in contact with the cap wafer.
 17. The method of claim 16 further comprising forming the cap wafer from Si with a trap-rich layer on a lower side of the cap wafer.
 18. The method of claim 16 further comprising forming the antenna terminal from an adhesion layer, a first metal layer disposed on the adhesion layer, and a second metal layer disposed on the first metal layer.
 19. The method of claim 16 further comprising forming a post on the reception signal terminal extending from the reception signal terminal to the cap wafer and forming a post on the transmit signal terminal and extending from the transmit signal terminal to the cap wafer.
 20. The method of claim 16 further comprising forming one or more ground terminals on the support substrate with posts disposed on the one or more ground terminals and extending from the one or more ground terminals to the cap wafer.
-